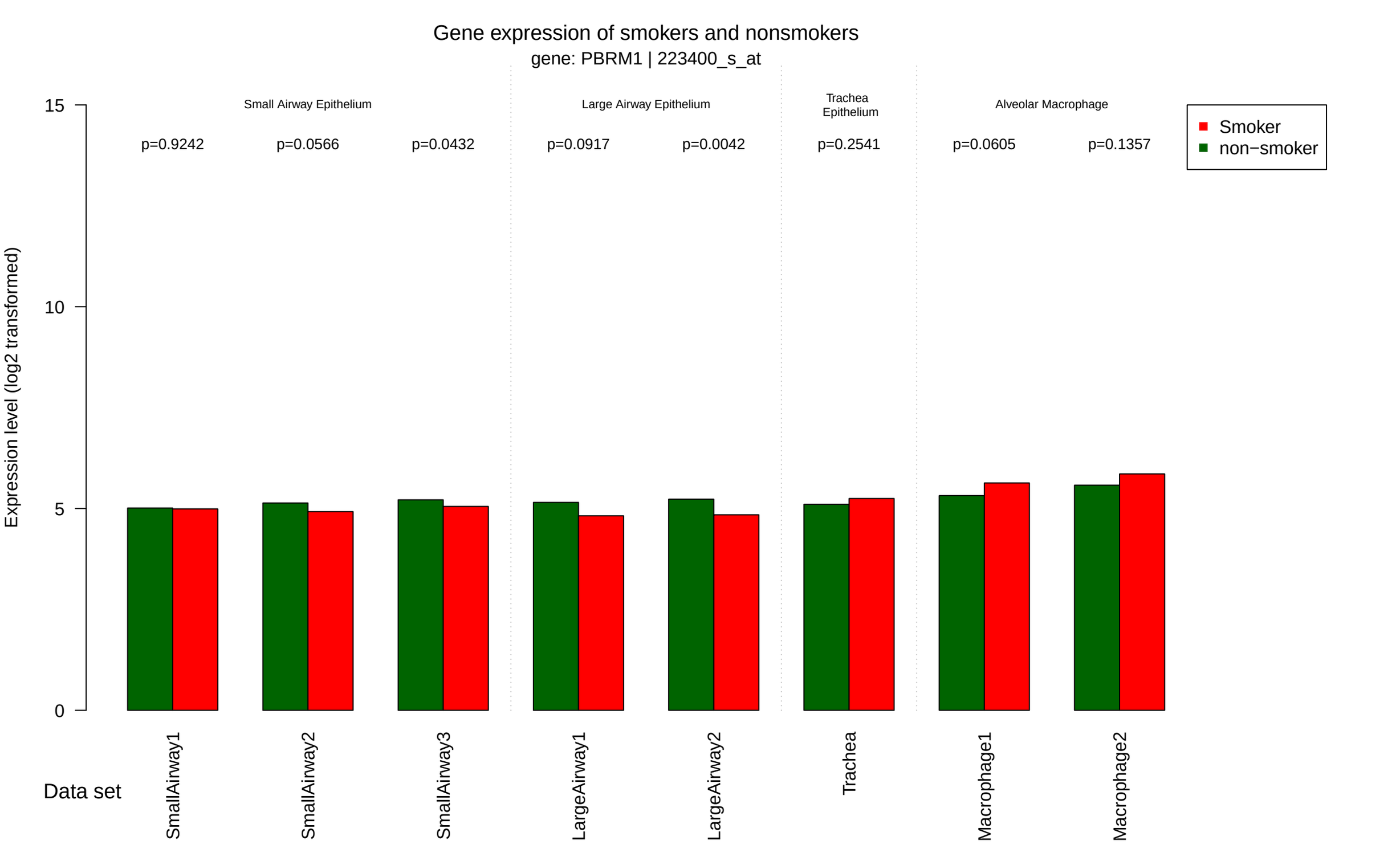
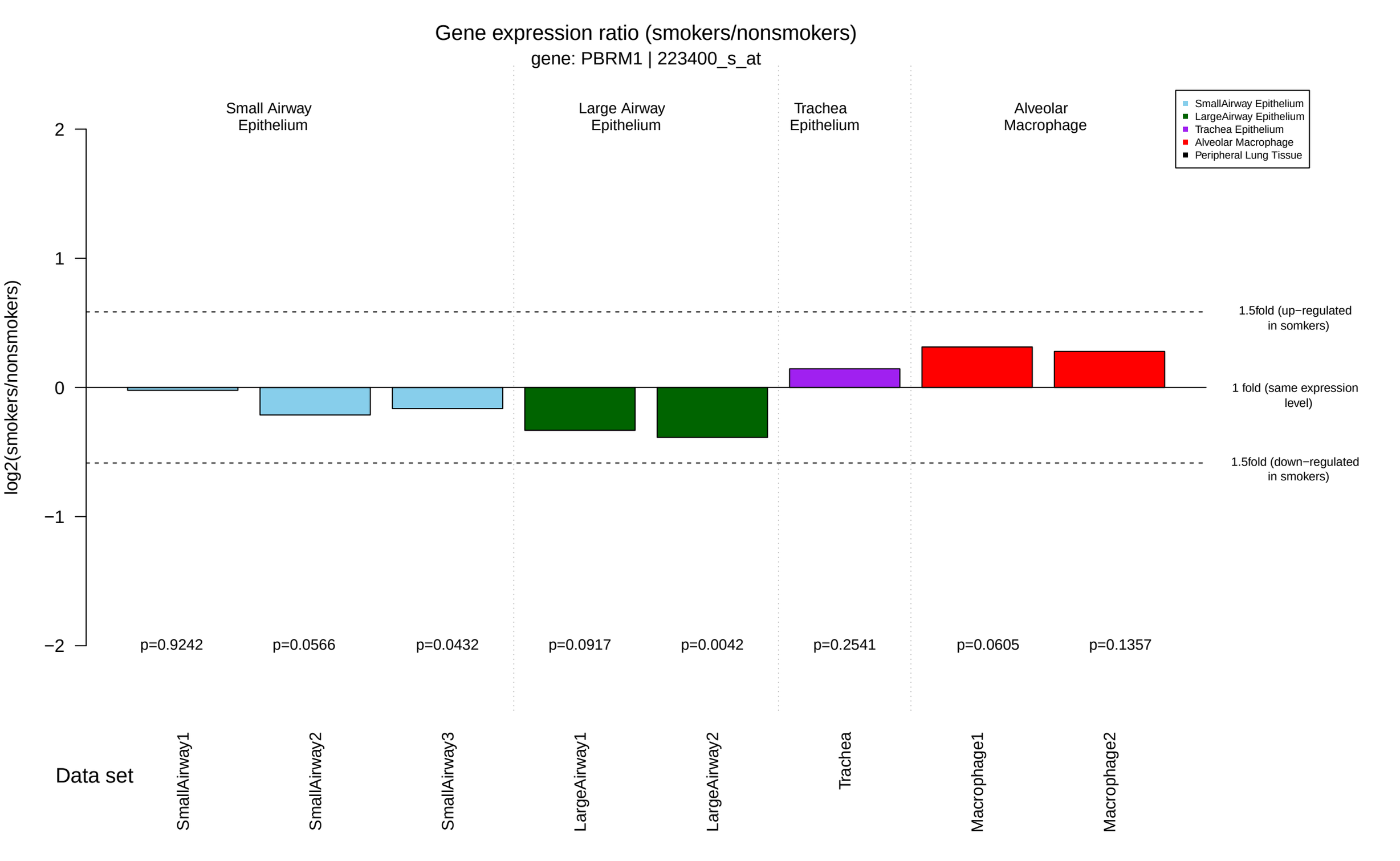
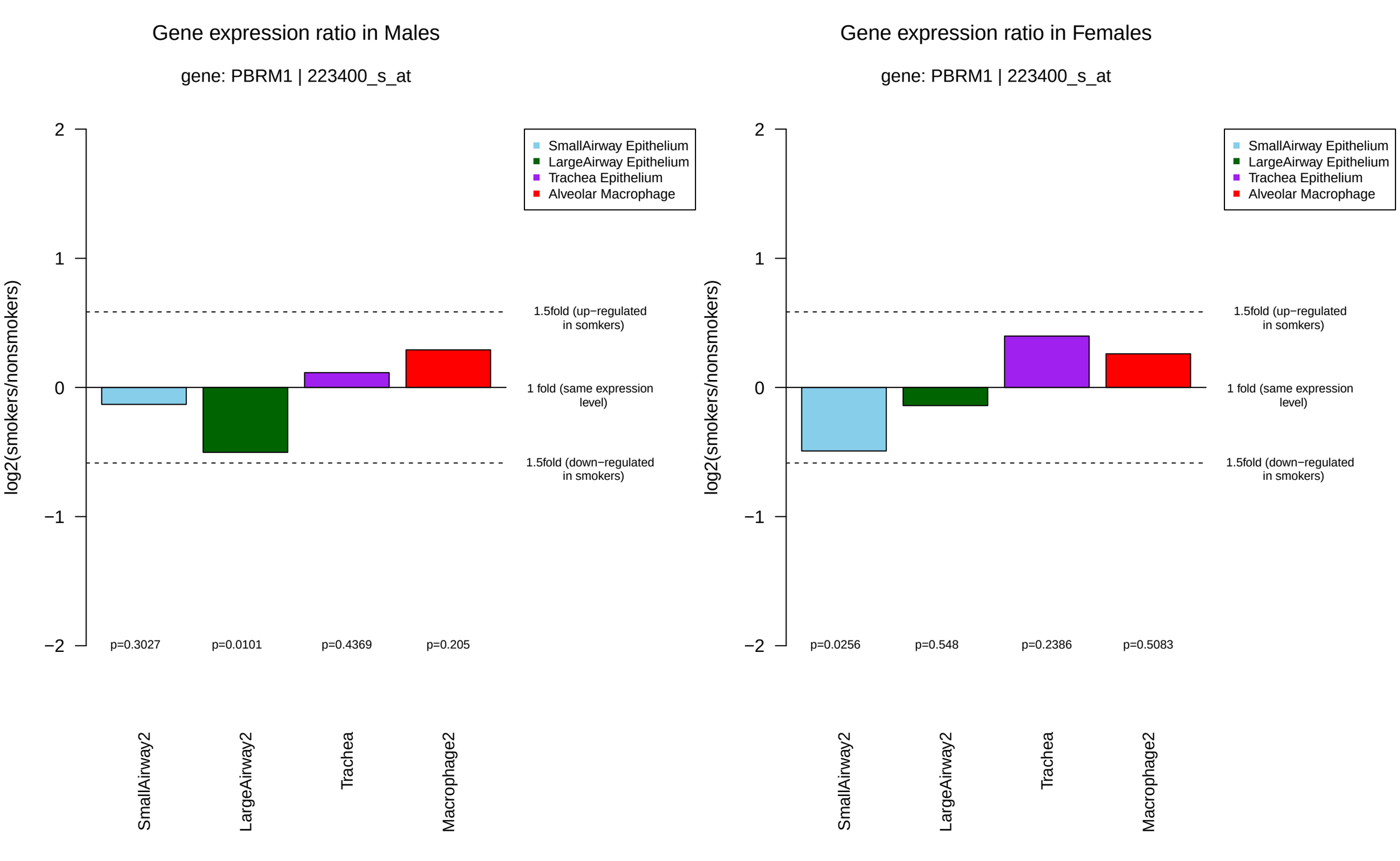
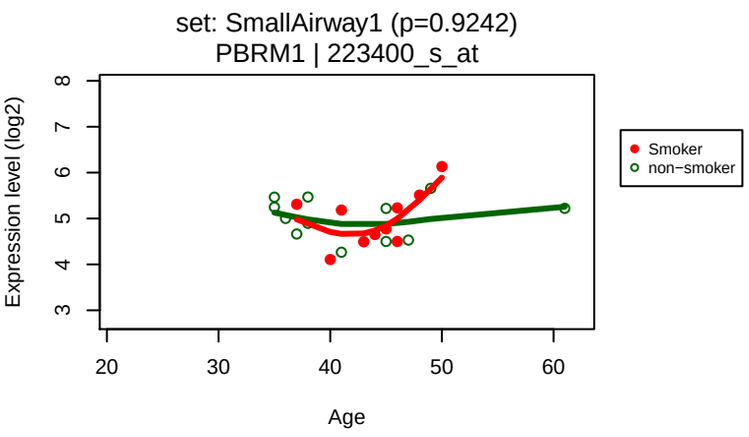
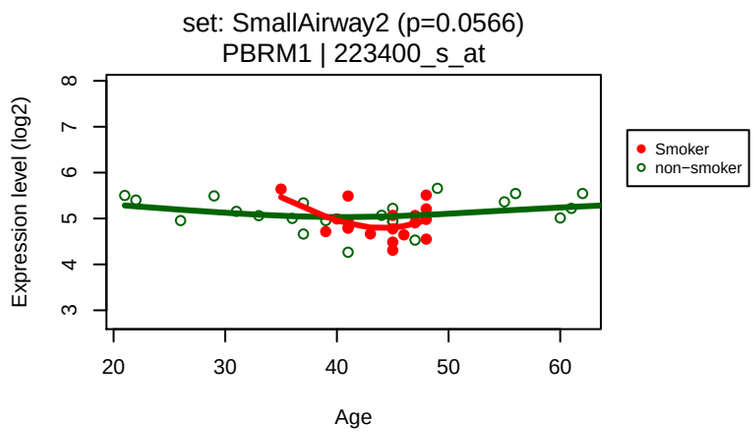
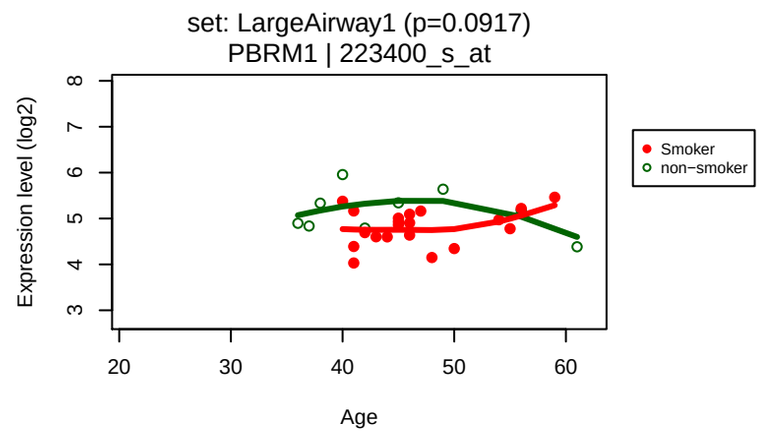

Gene expression of smokers and nonsmokers
gene: PBRM1 | 223400_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9242
p=0.0566
p=0.0432
p=0.0917
p=0.0042
p=0.2541
p=0.0605
p=0.1357
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

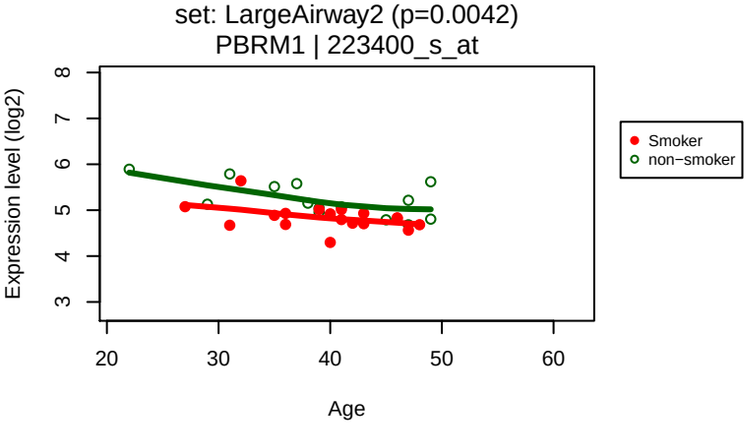

Gene expression ratio (smokers/nonsmokers)
gene: PBRM1 | 223400_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9242
p=0.0566
p=0.0432
p=0.0917
p=0.0042
p=0.2541
p=0.0605
p=0.1357
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

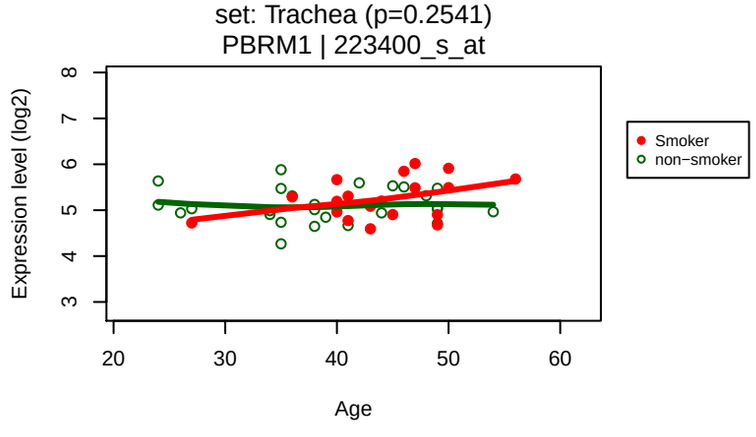

Gene expression ratio in Males
gene: PBRM1 | 223400_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3027
p=0.0101
p=0.4369
p=0.205
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PBRM1 | 223400_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0256
p=0.548
p=0.2386
p=0.5083
SmallAirway2
LargeAirway2
Trachea
Macrophage2

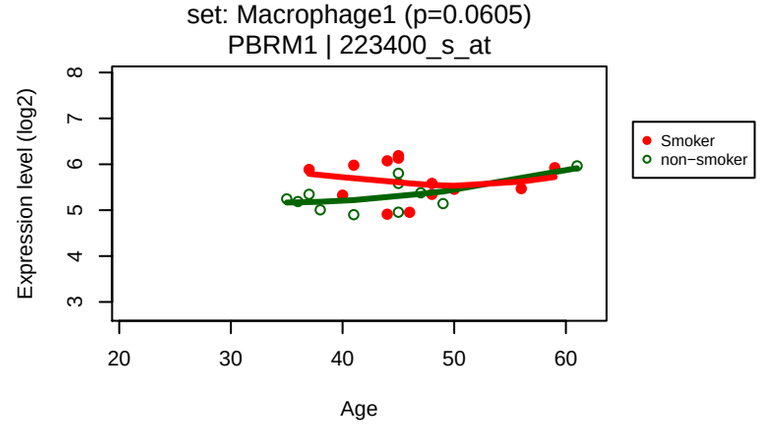

set: SmallAirway1 (p=0.9242)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker

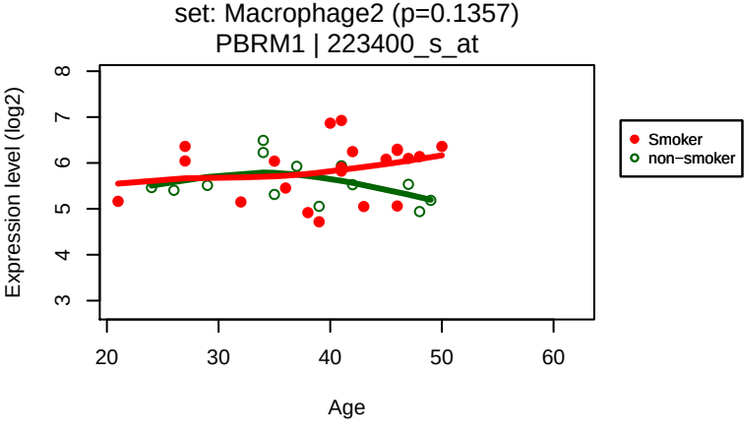

set: SmallAirway2 (p=0.0566)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker

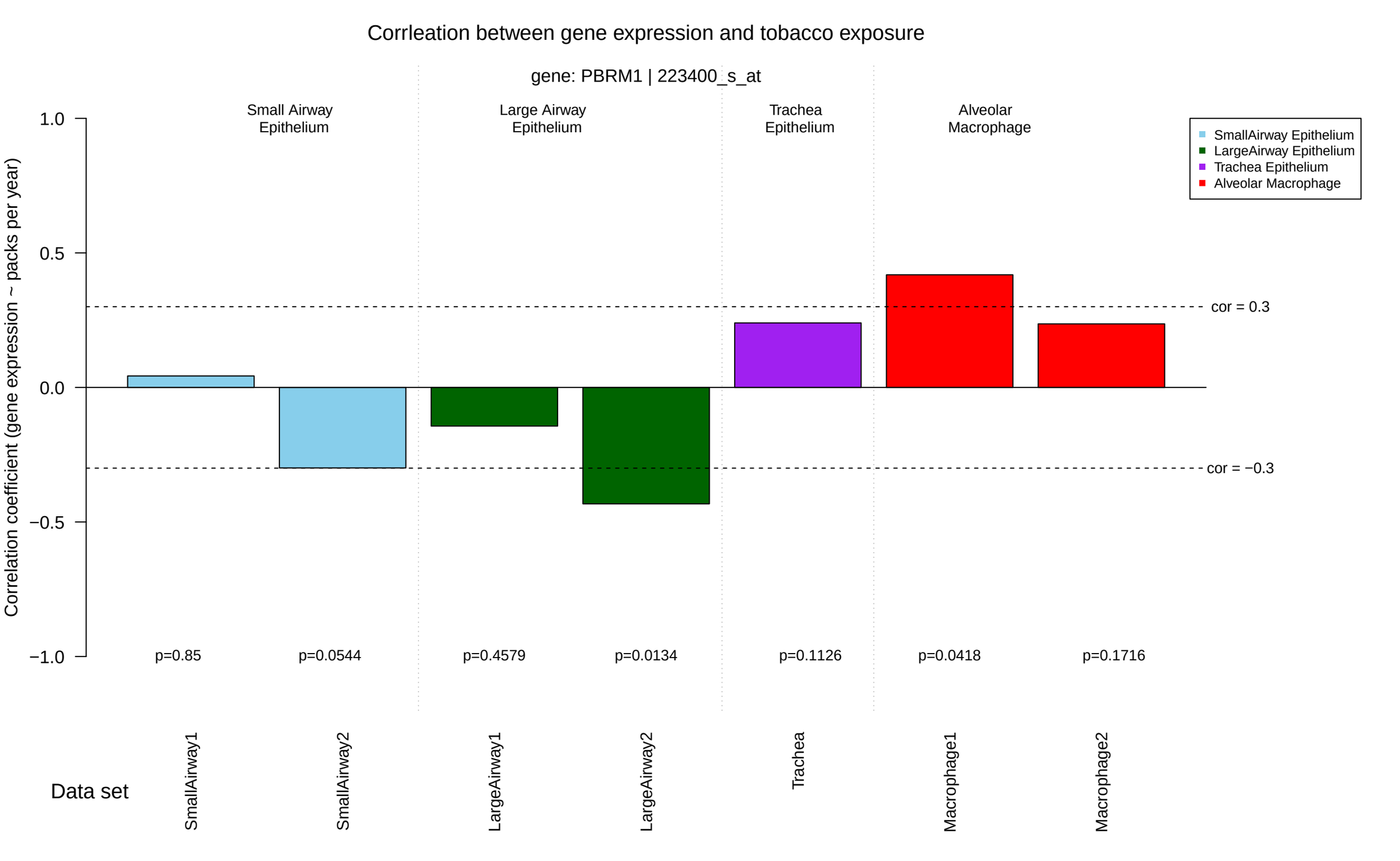

set: LargeAirway1 (p=0.0917)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0042)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2541)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0605)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1357)
PBRM1 | 223400_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PBRM1 | 223400_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.85
p=0.0544
p=0.4579
p=0.0134
p=0.1126
p=0.0418
p=0.1716
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2